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SOFTWARE PROJECT

FoodieNetwork

Cooking Lovers Application

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Chosen technology stack

For the cooking lovers application we will be using Html, Css, JavaScript and Django(Python).

HTML

HTML, or HyperText Markup Language, is the foundational language used to create web pages. It provides the structure and content of a webpage, allowing browsers to render and display text, images, and other media in a user-friendly format.

HTML elements are the building blocks of HTML pages. They are represented by a start tag, some content, and an end tag.

Web browsers are designed to read HTML documents and render them properly. While browsers do not display the HTML tags themselves, they use the tags to interpret how to display the content of the page.

HTML has evolved over the years, with HTML5 being the latest standard, ensuring modern features and optimizations are included for the best user experience.

In summary, HTML is a crucial language for web development, allowing the creation of structured, accessible, and interactive web pages. As you delve deeper into HTML, you'll learn about more complex elements and attributes that enable you to build rich and engaging web content

CSS

CSS (Cascading Style Sheets) is a style sheet language used to describe the presentation of a document written in HTML or XML. It is a cornerstone technology of the World Wide Web, alongside HTML and JavaScript

CSS allows you to control the layout, colours, fonts, and overall design of web pages. It enables the separation of content and presentation, which improves content accessibility and provides more flexibility and control over the presentation characteristics. This separation also allows multiple web pages to share formatting by specifying the relevant CSS in a separate .css file, reducing complexity and repetition in the structural content.

CSS is a rule-based language where you define rules by specifying groups of styles that should be applied to particular elements or groups of elements on your web page. A CSS rule consists of a selector and a declaration block. The selector points to the HTML element you want to style, and the declaration block contains one or more declarations separated by semicolons. Each declaration includes a CSS property name and a value, separated by a colon.

CSS is an essential technology for web development, providing the tools needed to create visually appealing and accessible web pages. By separating content from presentation, CSS allows for more efficient and flexible web design

JAVASCRIPT

JavaScript is a scripting language that enables you to create dynamically updating content, control multimedia, animate images, etc.

It plays a crucial role in both front-end and back-end development, enabling dynamic and interactive user experiences. Unlike compiled languages, JavaScript is interpreted through web browsers, making it highly accessible and versatile. It is a loosely typed (or weakly typed) language, meaning variable types are determined at runtime.

It is used together with HTML, CSS to create dynamic and functional web pages or websites.

FIGMA

Figma is a powerful design tool that lets you create and collaborate on user interfaces and prototypes for websites, apps, and much more. It is versatile and user-friendly.

DJANGO

Django is a high-level Python web framework framework that enables rapid development of secure and maintainable websites. It follows the "batteries included" philosophy, providing almost everything developers might need out of the box. Django is free and open-source

Django can be used to build almost any type of website, from content management systems and wikis to social networks and news sites. It supports various client-side frameworks and can deliver content in multiple formats, including HTML, RSS feeds, JSON, and XML.

It is :-

- Secure - It provides a secure way to manage user accounts and passwords. Helps developers avoid common security mistakes by providing a framework that automatically protects against vulnerabilities like SQL injection, cross-site scripting, cross-site request forgery, and clickjacking
- Scalable - Uses a component-based "shared-nothing" architecture, allowing it to scale for increased traffic by adding hardware at any level, such as caching servers, database servers, or application servers.
- Maintainable - promotes the creation of maintainable and reusable code by following design principles and patterns like the Don't Repeat Yourself (DRY) principle. It encourages grouping related functionality into reusable "applications" and modules
- Portable - written in Python, which runs on many platforms, including Linux, Windows, and macOS. This means you are not tied to any particular server platform

Django is a powerful and versatile web framework that simplifies the development of secure and maintainable websites. Its comprehensive features, scalability, and strong community support make it an excellent choice for web development

Application architecture

Application architecture is the process of defining a structure that meets all the technical and operational requirements of an application. It's the blueprint that guides the system's design and delineates how its various components interact with each other.

Our cooking lovers application is called FoodieNetwork.

Frontend with HTML, CSS, JavaScript – for UI interaction with the application, rendering of webpages like the log in, sign up, profile, feed, post recipe.

- Displays data received like recipe details, user input capture like comments and likes and recipe posting, sending user requests via forms and buttons to backend.

Backend with Django(Python) - core of the application, processing user requests, managing data, and handling business logic.

- Define the structure of the database like user ,comment and like .Handle user requests, interact with models, and return appropriate responses. Render HTML pages dynamically.

Data flow - Users interact with the frontend like submitting a new recipe. JavaScript or form submission sends the data to the backend via HTTP requests.

The backend sends the processed data back to the frontend. The frontend updates the UI to reflect changes e.g., displaying a new recipe.

User-uploaded content (e.g., recipe images) is stored in a media folder and managed using Django.

Description of Application Functionality

The FoodieNetwork web application will feature a log in page for registered users and a sign up page for new users. It is also for a regular user role.

There is a community feed which displays followed accounts and their posted pictures and recipes. Here there is also popular food suggestion.

You can also search for specific recipes based on meal time, cuisine, holiday, special diets or a specific recipe and can search for other users too.

A user can like, comment and share other people's recipes on other social platforms or even share recipes of their own.

There is also an option for saving recipes to try later.

A user can generate a shopping list based on the recipe they want to try and check if there are any substitutes they can use.

A user can manage their account, deleting recipes, unfollowing other users and can set up profile pictures and security measures.

Description of User Interface

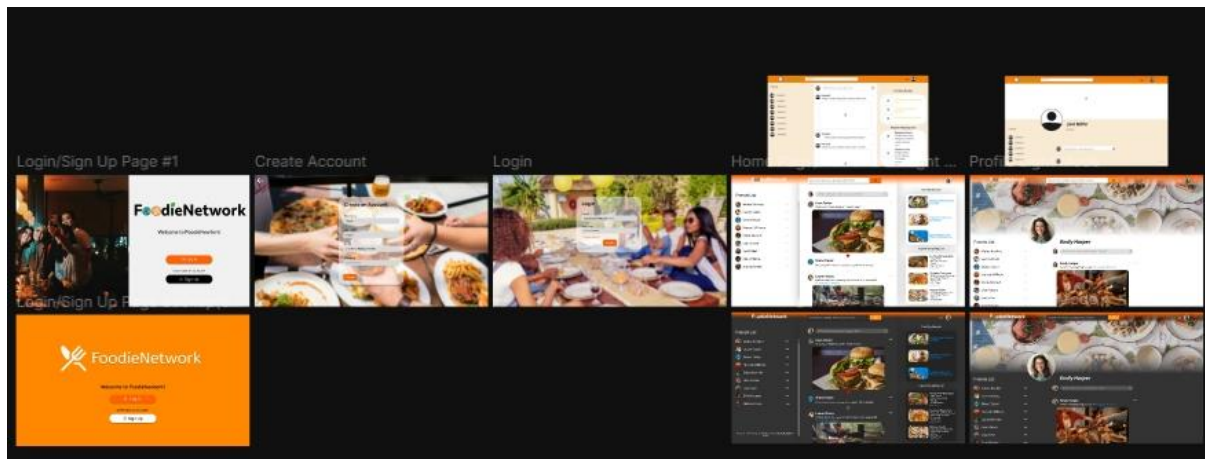
The FoodieNetwork web app has a clean and modern interface with an orange colour palette. Orange is an energetic colour that evokes feelings of excitement, enthusiasm and warmth that is why we chose it.

There is a consistent use of buttons, icons and layout for ease of use.

The key pages include :-

- Log in – a simple form for user authentication
- Sign up – a simple form for new user registration
- Homepage – recipe highlights and navigation
- Feed – features recipes posts from followed users and suggested posts
- Profile – user picture, user details, shared and saved recipes
- Recipe page – where a user can view a detailed recipe and also post their recipe.

The app will be developed with time to make it mobile friendly.



Summary

This platform is for cooking enthusiasts who like to share, explore and try new recipes and also express themselves.

HTML, CSS and JavaScript for 90% of our front end because it is easy to learn and understand, JavaScript adding the interactivity.

Figma is used for prototyping and getting a picture of what the web app should/could look like.

Django (Python) for the backend because it is a powerful and versatile web framework that simplifies the development of secure and maintainable websites.

The web app offers recipe posting and sharing and shopping list generation.

(Later on real-time notifications, video sharing, user analytics may be developed.)

SoftwareProject

(Sources- [w3school.com](https://www.w3school.com), docs.djangoproject.com, developer.mozilla.org,
[geeksforgeeks.org](https://www.geeksforgeeks.org) , javascript.info, [unsplash](https://unsplash.com), [figma](https://www.figma.com))